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# Explainable AI and Machine Learning for Transparent Building Energy Benchmarking and Retrofit Decisions

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## Abstract

Recent advances in artificial intelligence (AI) have driven broad application across industries, with machine learning systems now demonstrating superhuman performance in numerous tasks. However, this leap in performance has often come at the cost of increased model complexity, turning many systems into opaque "black boxes" and raising concerns about how decisions are made. This lack of transparency poses challenges for deploying AI in sensitive yet high-impact domains, such as energy benchmarking, where trust and accountability are crucial. Energy benchmarking in commercial buildings is often the first step toward evaluating performance and determining whether further interventions are needed. Yet, progressing beyond benchmarking remains challenging. Identifying buildings suitable for upgrades or pinpointing opportunities for energy savings and operational cost reduction is often hindered by the complexity and cost of in-depth assessments. To overcome these barriers, this study integrates data-driven modeling, machine learning and Explainable AI (XAI) to enhance both energy benchmarking and retrofit evaluation processes. Building stock-scale counterfactuals showed that envelope upgrades with high-performance HVAC achieved 9% average savings, while lighting modernization with advanced controls yielded 11% median savings.

## Introduction

Buildings account for approximately 30% of global energy use (International Energy Agency 2023), proving a tremendous opportunity for substantial energy reductions and cost savings for building owners. Energy benchmarking provides a critical foundation for evaluating the performance of existing buildings, enabling comparison against peers, identification of inefficiencies, and hence guiding the development of tailored actions for energy reduction. Data-driven benchmarking methods, including both statistical models and data analytics techniques, are powerful tools for characterizing building energy performance across various time scales (yearly, monthly, daily, hourly) and at increasing spatial levels of analysis, ranging from individual buildings to a group of buildings or a campus (Wei et al. 2018). The authors used a multilevel Bayesian model to analyze commercial building energy consumption in Europe, incorporating multiple parameters, including location and building type, to estimate energy consumption intensity for benchmarking. (Li et al. 2024) used real-world data from several cities in the United States.

However, benchmarking is simply the first step to realized energy savings. It helps assess building performance and benchmarks the same against a group of peers. However, it needs to be followed by detailed retrofit assessments to identify cost effective retrofit strategies for the desired target energy savings or a long term strategic capital investments. Retrofit assessments are typically carried out through on-site energy audits, the procedures for which are defined in ASHRAE Standard 211, Standard for Commercial Building Energy Audits (ASHRAE 2018). These on-site assessments are typically time-consuming and resource intensive, limiting its use and application at the wide spread level required for realizing long-term energy savings.

Machine learning (ML) is increasingly being used as the first step when going from benchmarking assessments to on-site energy audits. These ML algorithms have the potential to streamline the process of going from benchmarking to on-site energy audits and reduce the cost and effort associated with the detailed assessment by using real or synthetic datasets to identify retrofit opportunities and potential for energy savings. These ML techniques complements traditional physics-based simulation approaches and possess the ability to serve as a data-driven layer for early-stage screening and portfolio benchmarking, providing actionable insights into where detailed physics-based analyses should be applied.

This paper identifies the state-of-the-art in developments related to ML and retrofit assessments and proposes a new capability which leverages building characteristic data through ML and Explainable Artificial Intelligence (XAI) to identify retrofit potential and energy savings. This approach has been developed using energy audit data submitted through United States Department of Energy's (US DOE) Audit Template tool (Goel et al. 2022).

The approach centers on estimating primary energy demand for space heating, space cooling and domestic hot water in commercial buildings, using insights drawn from a large dataset of approximately 10,000 Audit Template buildings submitted to New York City (NYC). The performance of four machine learning algorithms is evaluated to determine the most reliable predictive model. The best (trained) model is then used to assess current building energy performance and simulate post-retrofit scenarios. The XAI layer provides post-hoc explanations for these predictions, highlighting the key factors that drive the energy consumption at the level of individual buildings, acknowledging that these factors can vary substantially across buildings in real-world settings. Once identified, the model is applied to simulate post-retrofit scenarios, allowing for a comparative assessment of energy performance before and after the retrofits.

The key contributions of this paper are:

- Utilizing high-fidelity NYC Audit Template data to model complex commercial systems (e.g., VAVs, lighting controls) often absent in related studies.
- From Explanation to Quantification: Integrating SHAP with counterfactual simulations to directly estimate energy savings from targeted retrofits, moving beyond simple feature importance.
- Portfolio-Scale Decision Support: Demonstrating how XAI frameworks can support city-wide mandates (e.g., Local Law 87) by aggregating retrofit potential across diverse building stocks.

The rest of the paper is organized as follows. The **Background** section first outlines existing approaches and tools used for benchmarking and retrofit assessments. It then highlights how machine learning and XAI are enabling new ways to predict and analyze energy consumption, while also addressing the challenges that limit their use despite clear benefits. The **Methodology** section details the process for model development, feature attribution, and counterfactual analysis. The **Evaluation** section outlines the experimental setup, including data pre-processing and hyperparameter tuning. Finally, the **Results** section presents the outcomes of the XAI model at both the individual building and portfolio levels.

## Background

### Building Performance Benchmarking and Retrofit Assessment in Industry

Benchmarking and energy prediction serve complementary roles in evaluating and managing building performance. Benchmarking provides a comparative lens by positioning a building's energy use against historical performance, peer buildings, or normative standards after adjusting for factors such as weather and occupancy. Prediction, on the other hand, establishes expected future consumption, often utilized for new building design, retrofit planning, and meeting energy efficiency program requirements. Together, they support the implementation of performance requirements for individual buildings as well as portfolio-level management strategies.

Benchmarking has been institutionalized through tools such as the U.S. Environmental Protection Agency's *ENERGY STAR Portfolio Manager*, which applies multivariate linear regression with weather and operational characteristics normalization to generate 1–100 scores for millions of commercial buildings (U.S. Environmental Protection Agency 2023). Evaluations of energy efficiency programs show that mandatory benchmarking is associated with 3–8% reductions in average energy intensity over two to three years, with NYC reporting a 14% decline within four years of program implementation (U.S. Department of Energy 2019). Several benchmarking products have been available to the industry, such as Energy Star, Building Energy Quotient, EnergyIQ, Home Energy Yardstick, and Home Energy Scoring Tool, as reviewed by Lei et al. (Lei et al. 2015). These products provide benchmarking based on reference databases such as Commercial Buildings Energy Consumption Survey (CBECS), Commercial End-Use Survey (CEUS) and Residential Energy Consumption Survey (RECS). While traditional regression models and physics-based models remain dominant, recently, academia and industry has started experimenting with more advanced and non-linear machine learning approaches, such as gradient boosted trees with interpretability techniques, have demonstrated improved benchmarking accuracy and stakeholder trust (Arjunan, Poolla, and Miller 2020). Emerging trends point toward greater automation and program-driven demand for more rigorous, explainable methods. Machine learning enhancements, paired with feature attribution methods, are making benchmarking more actionable for building owners and regulators (Arjunan, Poolla, and Miller 2020).

Studies on retrofit assessment approaches have emphasized the predominance of energy-focused evaluations in retrofit research. (Dabous and Hosny 2025) in their systematic review of envelope retrofit strategies showed that most studies evaluate interventions primarily through modeled energy savings and thermal performance outcomes. (Toosi et al. 2020) highlighted the importance

of combining economic, and other pertinent dimensions over the building life. They noted that assumptions regarding system boundaries, time horizons, and data availability significantly influence assessment outcomes, contributing to inconsistencies across the literature. (Gonzalez-Caceres, Rabani, and Martínez 2019) examined residential retrofit tools and found that many of these tools emphasize energy and financial assessments, often using scenario analysis and simulation models. Together, these studies indicate that while retrofit assessment approaches are expanding in scope, they remain dominated by energy and cost considerations.

**Machine Learning and Explainable Artificial Intelligence (XAI) in Building Performance Evaluation**

A growing body of research has demonstrated ML's ability to predict whole-building and subsystem energy consumption, forecast loads, and support retrofit decision making with higher computation efficiency and flexibility than conventional simulation approaches (Kalaimani et al. 2020; Jain et al. 2019; Jain et al. 2021; Kömürcü and Edis 2025). However, despite demonstrated benefits, interpretability of complex models such as deep neural networks is a persistent concern, particularly when used to guide operational or investment decisions.

Explainable AI (XAI) provides methods to explain the decisions of AI models, making them more transparent and trustworthy (Darvishvand et al. 2025; Jain et al. 2023). The ability to explain *why* a model is projecting/estimating something helps users reason through model estimations and build trust in AI-systems, while also allowing developers to detect errors, uncover biases, and refine models. Among all, SHAP is the most widely adopted post-hoc method because it is model-agnostic and provides both global and post-hoc explainability grounded in cooperative game theory.

Research on XAI in buildings spans predictive modeling (Joshi et al. 2022), energy monitoring (Henkel et al. 2024), fault detection (Prabhu, Valdi, and Arjunan 2023), and optimization (Haghighat, MohammadiSavadkoohi, and Shafiabady 2025). Most studies train ensemble tree models or neural networks to estimate energy consumption and apply post-hoc techniques (such as SHAP) to identify key drivers. For instance, the XENIA framework benchmarks consumption using attributes such as age, floor area, lighting, and occupancy, with SHAP and PDP plots showing that AC floor area exerts the largest positive effect on energy use (Joshi et al. 2022).

While XAI has been applied to a variety of building-related tasks, few studies address retrofitting directly. (Wenninger et al. 2022) analyzed Energy Performance Certificate (EPC) data from England and Wales with socio-demographic and housing prices to predict retrofit uptake, showing house value, income, and regional programs as key factors. (Rempi et al. 2025) addressed retrofit dataset scarcity with Conditional Tabular GANs (for generating synthetic samples) and SHAP, improving classification by up to 54% and exposing spurious postcode effects. (Ye et al. 2024) scaled analysis to neighborhoods, combining parametric simulations, XGBoost surrogates, and multi-objective optimiation, with SHAP showing that PV benefits depended on floor-area ratio, while lighting upgrades were consistently beneficial.

Closest to our work, (Piscitelli et al. 2025) focused on advancing residential energy benchmarking using 49,000 EPCs from Italy's Piedmont region, applying ML with a two-layer XAI framework: one to ensure robustness by filtering unstable input regions, and another to generate local explanations for pre- and post-retrofit scenarios, ultimately enabling transparent benchmarking and comparative retrofit analysis. Our study extends this work to Audit Template data, covering thousands of commercial buildings subject to NYC's Local Law 87, aiming to uncover insights from this building stock and evaluate the outcomes for benchmarking and retrofit assessment.

## METHODOLOGY

Our approach uses the Audit Template data and integrates supervised ML with XAI to highlight key drivers behind commercial building energy performance and estimate the impact of retrofit measures. The NYC dataset (City of New York 2021) provides a comprehensive resource for empirical research and analysis. Covering audits conducted between 2019 and 2024 and submitted through Audit Template tool, the dataset encompasses detailed building-level information on the envelope, lighting, HVAC, and service hot water systems. It also contains reported utility consumption, annual energy use disaggregated by fuel type and end use, and documented energy efficiency measures, including measure costs and projected savings. With $> 10{,}000$ building records, this dataset is used for model development.

The proposed workflow proceeds in three stages: (a) model development, (b) identification of key drivers, and (c) counterfactual simulation of retrofit scenarios.

**Model Development**

We first trained a suite of supervised machine learning algorithms to predict building-level annual site energy use

intensity (EUI). We consider a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^p$ represents standardized building characteristics, equipment specifications, and operational parameters for building $i$, and $y_i \in \mathbb{R}$ denotes its observed site EUI. A suite of supervised machine learning algorithms $\{f_m\}_{m=1}^{M}$ was trained to learn the mapping $\mathbf{x}_i \mapsto y_i$. Algorithms included linear models, tree-based learners, and ensemble approaches, each selected to capture different types of relationships between building features and energy outcomes. Linear models are interpretable and serve as a baseline for additive effects, whereas tree-based and ensemble methods are better suited for nonlinear interactions and heterogeneous feature spaces. All input features were standardized, categorical variables were encoded, and missing values were imputed prior to training to ensure comparability across algorithms.

For every algorithm $f_m(\,;\theta_m)$, the optimal parameters were learned by minimizing the empirical loss with an added regularization term. Here, $\theta_m$ denotes the set of model parameters and hyperparameters that are estimated during training.

$$\hat{\theta}_m = \arg\min_{\theta_m} \frac{1}{N} \sum_{i=1}^{N} L(y_i, f_m(\mathbf{x}_i;\theta_m)) + \lambda_m R(\theta_m); \quad (1)$$

- $N$ = number of training samples,
- $\mathbf{x}_i$ = feature vector of sample $i$,
- $y_i$ = target response for sample $i$,
- $L(\,,\,)$ = loss function (mean squared error),
- $R(\,)$ = model-specific regularization term,
- $\lambda_m$ = regularization penalty weight.

Hyperparameters $\theta_m$ and $\lambda_m$ were optimized via $K$-fold cross-validation to ensure robust out-of-sample performance. The use of multiple algorithms allows us to assess the stability of predictive performance and subsequent post-hoc analyses across model classes, thereby reducing dependence on a single functional form.

**Feature Attribution with SHAP**

Understanding which factors drive energy consumption is essential for prioritizing retrofit opportunities and guiding program interventions. To deconstruct the primary drivers of building energy use, we applied SHapley Additive exPlanations (SHAP) (Lundberg and Lee 2017) to the best-performing predictive model, as identified through cross-validation. SHAP is based on the concept of Shapley values from cooperative game theory, where the prediction of a model for an input $\mathbf{x}_i$ is decomposed into additive contributions of individual features. For a model $f$ and instance $\mathbf{x}_i$, the prediction can be expressed as

$$f(\mathbf{x}_i) = \phi_0^{(m)} + \sum_{j=1}^{p} \phi_j^{(m)}(\mathbf{x}_i); \quad (2)$$

where $\phi_0^{(m)}$ is the expected prediction across the dataset and $\phi_j^{(m)}(\mathbf{x}_i)$ denotes the marginal contribution of feature $j$ to the prediction for instance $i$. The value $\phi_j^{(m)}(\mathbf{x}_i)$ is computed by averaging the incremental contribution of feature $j$ over all possible subsets of features, ensuring properties of fairness and consistency. This additive decomposition provides both *global explanations*, obtained by aggregating $|\phi_j^{(m)}|$ across all buildings, and *local explanations*, obtained at the individual building level. The best model and its SHAP-based attributions were subsequently used for the counterfactual retrofit analysis.

**Counterfactual Retrofit Analysis**

For features identified as critical by SHAP, counterfactual values were assigned to reflect retrofit scenarios. The reader must note here that while SHAP provides consistent additive attributions for a fixed trained model, the resulting explanations remain correlational and can be sensitive to feature correlations and model uncertainty. Therefore, SHAP outputs are only used to prioritize candidate drivers for further analysis rather than to infer causal retrofit effects.

Let $\mathbf{x}_i^{\prime} = g(\mathbf{x}_i)$ denote the counterfactual input vector obtained by applying a retrofit transformation $g(\,)$, such as reducing infiltration rates, upgrading HVAC efficiencies, or modifying lighting coverage shares (i.e., the proportion of floor area served by each lighting technology).

Retrofit transformations were grounded in plausible engineering ranges, for example by replacing low-efficiency equipment with code-compliant high-efficiency alternatives or adjusting envelope parameters within values observed in best-practice retrofits. Both single-feature interventions and combined multi-feature scenarios were considered, allowing the framework to capture interactions between retrofit measures. The best-performing model $f$ was re-evaluated with these adjusted inputs, and the retrofit impact was computed as

$$\Delta y_i = f(\mathbf{x}_i) - f(\mathbf{x}_i^{\prime}); \quad (3)$$

where $\Delta y_i$ represents the estimated reduction in site EUI

for building $i$.

The integration of retrofit transformations within the same predictive model allows direct quantification of potential savings under both individual and bundled interventions. This systematic approach not only isolates the technical levers that most influence building energy consumption but also produces actionable estimates of retrofit impacts, enabling prioritization of measures at the building and portfolio scale.

## EVALUATION

The proposed methodology was evaluated on the Audit Template dataset for NYC, which provides detailed records of building characteristics, equipment specifications, and reported energy performance. In this section, we will assess three aspects: (i) the predictive accuracy of different supervised learning algorithms in estimating site EUI, (ii) the stability and explainability of feature attributions produced by SHAP, and (iii) the reliability of counterfactual simulations in quantifying the potential impacts of retrofit measures. By jointly examining these dimensions, the evaluation aims to demonstrate both the predictive strength of the models and the practical utility of the framework in guiding retrofit decision-making.

### Experimental Setup

All analyses were conducted in Python 3.13. Linear, tree-based, and ensemble models were implemented using `xgboost`, `scikit-learn`, and `catboost`. Hyperparameters for each model were tuned via randomized search with $K$-fold cross-validation, using minimization of the cross-validated mean squared error on the *log-transformed* target as the objective. The best configuration for each model was refit on the training set, evaluated in log space, and additionally reported in the original target space via inverse transformation for explainability. All experiments were run on a MacBook Pro with an Apple M3 Max chip, equipped with 16 CPU cores and a 40-core integrated GPU. Machine learning models were executed on CPU. To ensure consistency across runs, random seeds were fixed for all libraries.

### Dataset

For evaluation, we utilized the energy audit dataset for NYC commercial buildings, submitted through Audit Template (City of New York 2021). Audit Template is a web-based tool for collecting and storing existing building asset and performance data (Goel et al. 2022). It is used by over 15 states, localities and programs for implementation of their energy efficiency programs. These programs require compliance with ASHRAE Standard 211, Standard for Commercial Building Energy Audits (ASHRAE 2018) which defines procedures and reporting requirements for ASHRAE Standard Level 1, 2 and 3 energy audits. Over 2000 existing buildings are entered in the Audit Template every year. The data reported in Audit Template is stored in structured format, making it ideal for data-driven approaches and analysis frameworks. The version used in this study covers audits submitted between 2019 and 2024 and contains 10,702 building-level records with 1,992 reported variables. Each record corresponds to an energy audit filed under the NYC program, which requires properties larger than 50,000 square feet to undergo periodic audits and retro-commissioning. The dataset spans all five boroughs, with the largest representation from Manhattan ($n$ = 4527), followed by Queens ($n$ = 2106), Brooklyn ($n$ = 2097), the Bronx ($n$ = 1751), and Staten Island ($n$ = 205).

Reported attributes include property identifiers, location (borough, block, lot), year of construction (ranging from the 1800s to 2024), and total floor area (mean: 156,715 $ft^2$, max: 20,293,750 $ft^2$). System-level details are also documented, including heating fuel type, boiler and chiller capacity, cooling efficiency, ventilation systems, domestic hot water equipment, lighting types and coverage, and envelope characteristics. This combination of building characteristics and system specifications provides a comprehensive basis for modeling site energy use intensity (EUI) and assessing the potential impacts of retrofit measures.

### Data Pre-processing

The raw audit data were first cleaned by removing records with zero reported annual energy use or zero conditioned floor area, which result in undefined or implausible site EUI values. After this filtering, 7,263 valid building records remained.

A building-level feature matrix was then constructed, focusing on features that represent major building systems and retrofit opportunities, such as lighting, HVAC, envelope, heating, and cooling (Table 1). These groups capture the largest energy-consuming components in commercial buildings and therefore provide the most actionable levers for retrofit analysis. In total, this feature engineering produced 331 features.

Missing values were imputed according to feature type. For categorical and coverage-related features, missing entries were replaced with zero, indicating the absence of that system or component. For continuous numerical

Table 1: Summary of preprocessing steps and engineered features.

| Group | Feature | Notation | Formulation | Notes |
|---|---|---|---|---|
| Target | Site EUI | | $EUI_i = \frac{E_i^{\text{tot}}}{A_i^{\text{cond}}}$ | $E_i^{\text{tot}}$ = total annual energy use (kBtu), $A_i^{\text{cond}}$ = conditioned floor area (ft$^2$). Records with $EUI_i$ 0 removed. |
| Lighting | Coverage share | `Lighting_`*h*`Tech`*i*`_Cov` | $L_{ij} = \frac{A_{ij}}{\sum_{k=1}^{p} A_{ik}}$ | $A_{ij}$ = floor area served by lighting type $j$ in building $i$, $p$ = total lighting types. |
| | Controls | `LightingControlType_`*h*`Control`*i* | One-hot indicator | Encoded control categories (e.g., occupancy sensors). |
| | Number of fixtures | `LightingFixtures_`*h*`Tech`*i* | $N_i^{\text{fix}} = \sum_{j=1}^{p} F_{ij}$ | $F_{ij}$ = number of fixtures of type $j$ in building $i$, $p$ = total lighting types. |
| HVAC | Coverage share | `HVAC_`*h*`System`*i*`_Cov` | $H_{ij} = \frac{A_{ij}}{\sum_{k=1}^{q} A_{ik}}$ | $A_{ij}$ = floor area served by HVAC type $j$ in building $i$, $q$ = total HVAC types. |
| | Controls | `HVACControls_`*h*`Control`*i* | One-hot indicator | Encoded HVAC control categories (e.g., EMS, scheduling). |
| Roof | Avg. R-value | `Roof_R_mean` | $R_i^{\text{roof}} = \frac{1}{n_i} \sum_{j=1}^{n_i} R_{ij}$ | $R_{ij}$ = $R$-value of roof segment $j$, $n_i$ = number of roof segments. |
| | Roof type | `Roof_`*h*`Type`*i*`_I` | One-hot indicator | Encoded roof type categories reported in LL87. |
| Wall | Avg. R-value | `Wall_R_mean` | $R_i^{\text{wall}} = \frac{1}{m_i} \sum_{j=1}^{m_i} R_{ij}$ | $R_{ij}$ = $R$-value of wall segment $j$, $m_i$ = number of wall segments. |
| Window | Window-to-wall ratio | `Window to Wall Ratio` | $WWR_i$ | Given in the data. |
| | Glass type | `Roof_`*h*`Glaze`*i*`_I` | One-hot indicator | Encoded window glazing type categories. |
| Heating | Efficiency | `Heating_Efficiency_Avg` | $\eta_i^{\text{heat}} = \frac{\sum_{j=1}^{m_i} \eta_{ij} C_{ij}}{\sum_{j=1}^{m_i} C_{ij}}$ | $\eta_{ij}$ = efficiency of heating unit $j$, capacity-weighted. |
| | Average age | `Heating_Age_Avg` | $A_i^{\text{heat}} = \frac{1}{m_i} \sum_{j=1}^{m_i} (Y \quad Y_{ij})$ | $Y$ = reference year (2025), $Y_{ij}$ = installation year of unit $j$. |
| | Heating source + Fuel type | `HS_`*h*`Source`*i* *h*`Unit/Desc`*i* *h*`Fuel`*i*`_I` | One-hot indicator | Encoded heating source with fuel type categories. E.g., `HS_Plant_Steam Boiler_Natural Gas_I` depict that heating is coming from Plant with fuel type being natural gas. |
| Cooling | Efficiency | `Cooling_Efficiency_Avg` | $\eta_i^{\text{cool}} = \frac{\sum_{j=1}^{n_i} \eta_{ij} C_{ij}}{\sum_{j=1}^{n_i} C_{ij}}$ | $\eta_{ij}$ = efficiency of cooling unit $j$, capacity-weighted. |
| | Average age | `Cooling_Age_Avg` | $A_i^{\text{cool}} = \frac{1}{n_i} \sum_{j=1}^{n_i} (Y \quad Y_{ij})$ | $Y$ = reference year (2025), $Y_{ij}$ = installation year of unit $j$. |
| | Cooling source + Fuel type | `CS_`*h*`Source`*i* *h*`Unit/Desc`*i* *h*`Fuel`*i*`_I` | One-hot indicator | Encoded heating source with fuel type categories. E.g., `CS_Plant_Steam Boiler_Natural Gas_I` depict that heating is coming from Plant with fuel type being natural gas. |

features, missing values were imputed using the mean and then standardized using a standard scaler.

To stabilize variance and reduce skewness in the response variable, site EUI was log-transformed prior to modeling. Log transformation of energy outcomes has been shown to improve predictive performance in regression tasks by reducing the influence of extreme values. The transformed EUI was then imputed with its mean for records where the target was missing.

The processed dataset was randomly partitioned into training (70%) and testing (30%) sets. Hyperparameter tuning was performed on the training set using $K$-fold cross-validation, and model performance was subsequently evaluated on the held-out test set.

## Hyperparameter Tuning

Hyperparameter tuning was performed using `RandomizedSearchCV` with $n_{\text{iter}} = 30$ and 5-fold cross-validation (shuffled, fixed seed). Random Forest, XGBoost, and CatBoost were tuned, while Linear Regression was retained as an untuned baseline. Model selection and evaluation were based on RMSE in the log-transformed target space. Among the tuned models, CatBoost achieved the strongest performance with a test RMSE of 1.04, MAE of 0.49, and $R^2$ of 0.54 in log space, using 2000 boosting iterations, a learning rate of 0.03, depth of 10, and $l_2$ regularization of 3. Random Forest and XGBoost followed closely with test RMSE values of 1.07 and 1.08, respectively, while Linear Regression performed worse with a test RMSE of 1.35. Table 2 summarizes the hyperparameter search ranges, with selected values highlighted in **bold**, together with the corresponding test performance metrics.

## Model Performance Evaluation

Hyperparameter tuning results showed that the CatBoostRegressor provided the best predictive accuracy among all candidate models. Next, we retrained the model on the complete training dataset using the selected configuration. On the held-out test set, this final model achieved a log-space RMSE of 1.04, MAE of 0.49, and $R^2$ of 0.54, confirming its strong generalization ability. Given its superior accuracy, CatBoost Regressor is adopted for the subsequent analyses, including SHAP-based explanation of feature importance and counterfactual evaluation of retrofit strategies.

Table 2: Hyperparameter search spaces and test performance (log space). Selected parameters are shown in **bold**.

| Model | Linear Regression | Random Forest | XGBoost | CatBoost |
|---|---|---|---|---|
| **Hyperparameters (search space)** | No tuning (baseline) | `n_estimators` = {300; **600**; 1000; 1400}<br>`max_depth` = {**None**; 10; 20; 30}<br>`min_samples_split` = {**2**; 5; 10}<br>`min_samples_leaf` = {**1**; 2; 4}<br>`max_features` = {"sqrt"; "log2"; **0.5**; 0.8} | `n_estimators` = {600; 1000; **1400**}<br>`max_depth` = {4; 6; **8**; 10}<br>`learning_rate` = {**0.01**; 0.03; 0.05; 0.1}<br>`subsample` = {0.7; 0.8; **0.9**; 1.0}<br>`colsample_bytree` = {**0.6**; 0.8; 1.0}<br>`min_child_weight` = {1; **3**; 5; 7}<br>`reg_lambda` = {**0.0**; 0.5; 1.0; 2.0} | `iterations` = {1000; 1500; **2000**}<br>`learning_rate` = {0.01; **0.03**; 0.05}<br>`depth` = {6; **10**; 12}<br>`l2_leaf_reg` = {1; **3**; 5; 7} |
| **Test RMSE** | 1.35 | 1.07 | 1.08 | **1.04** |
| **Test MAE** | 0.69 | **0.47** | 0.52 | **0.49** |
| **Test $R^2$** | 0.231 | 0.515 | 0.513 | **0.542** |

# RESULTS

The results of the modeling and post-hoc analyses are presented in this section. First, SHAP values are used to provide a feature-level explanation of model predictions at both global and local scales. Second, counterfactual analysis is carried out to explore the impact of targeted retrofit strategies on predicted EUI.

## SHAP-based Model Explanation

To better understand the drivers behind predicted EUI, we applied SHAP analysis to the final CatBoost model. SHAP values provide a consistent framework for attributing the contribution of each feature to the model's predictions. Across the evaluated model classes, the top contributing feature groups were consistent across models, indicating robustness of the identified drivers.

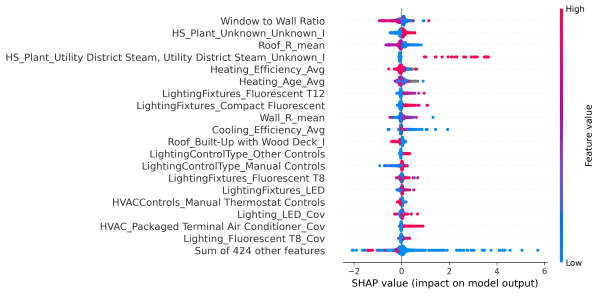



Figure 1: SHAP beeswarm plot for the CatBoost model

We used SHAP beeswarm plots to provide a compact summary of both the importance and direction of each feature's contribution to the prediction. In these plots, each point corresponds to one building (observation). The horizontal axis shows the SHAP value, i.e., the marginal impact of that feature on the model output. Points to the right indicate positive contributions (increasing predicted EUI), while points to the left indicate negative contributions (decreasing predicted EUI). The color encodes the raw feature value, ranging from low (blue) to high (red), allowing the analyst to see whether high or low values of a given feature drive predictions upward or downward. Features are listed vertically, sorted by an importance criterion.

Figure 1 shows the SHAP beeswarm plot for the CatBoost model. The plot orders features by their mean absolute SHAP values, highlighting the variables that contribute the most on average across all predictions. For instance, the left plot shows that lighting technologies (in particular, LED and fluorescent types), envelope characteristics (roof and wall thermal resistance), and WWRs are dominant factors. The color gradients further show that, for example, higher window-to-wall ratios (red) tend to increase EUI, while higher roof and wall insulation levels (red) reduce it, consistent with expectations.

Next, we used these localized insights as the basis for identifying retrofit opportunities.

## Counterfactual Analysis of Retrofit Strategies

To move beyond feature attribution and evaluate actionable interventions, we performed counterfactual analysis on selected high-EUI buildings. Counterfactuals are generated by perturbing input features in ways consistent with plausible retrofit actions, such as improving HVAC efficiency, upgrading insulation, or adjusting lighting technologies. Such comparisons highlight not only the magnitude of potential savings but also the heterogeneity of retrofit impacts across the building stock. We carry out the counterfactual analysis at two levels: (i) the building-level, where we identify and upgrade the most inefficient components, and (ii) the portfolio/building-stock level, where we simulate stock-wide retrofits and assess aggregate energy impacts.

## Building-Level Energy Impacts

At the building level, we use SHAP to identify the top three features most strongly associated with elevated

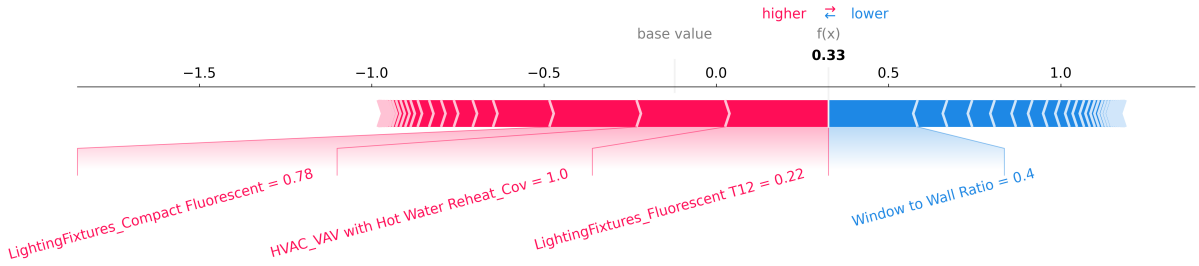


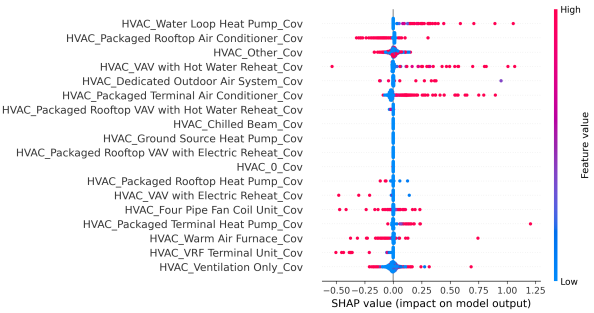

Figure 2: SHAP force plot depicting key factors contributing to the energy prediction in a representative building.

EUI. We hypothesise that these features are primary inefficiencies specific to each building and therefore serve as natural candidates for counterfactual intervention.

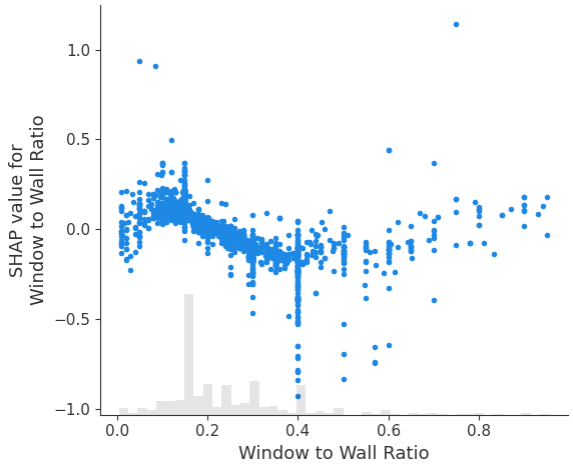
Figure 2, for example, shows the SHAP force plot for one representative building. The plot shows that the presence of inefficient fluorescent lighting (T12 and Compact) is the strongest contributor driving the prediction above the baseline, followed by HVAC_VAV with Hot Water Reheat_Cov, while the WWR of 0.4 (blue) pulls the prediction downward, contributing in lowering EUI.

Figure 3 further contextualizes these inefficiencies. The beeswarm plot (see Figure 3a), for instance, highlights that reliance on VAV with hot water reheat raises energy use in multiple cases, suggesting that substituting with packaged rooftop air conditioners, which are correlated with lower predicted EUI in the data, represents a plausible candidate for further evaluation rather than a causal recommendation. In fact, for the representative building in Figure 2, the annual ground truth EUI was 134 kBtu ft $^{2}$ yr $^{1}$ and the model predicted 140 kBtu ft $^{2}$ yr $^{1}$ pre-retrofit and 98 kBtu ft $^{2}$ yr $^{1}$ post-retrofitting, reflecting an annual savings of nearly 30%. Similarly, the dependence plot (on the right) shows a U-shaped relationship between window-to-wall ratio (WWR) and EUI, where both very low and very high values are associated with higher energy use. This swing in performance reflects the fact that WWR on its own does not dictate building energy consumption: the effect of window area is strongly mediated by other factors such as lighting demand, HVAC system efficiency, and the presence of advanced controls (see Figure 3b).

Next, we extend the analysis from a single building to the whole NYC dataset, using the same SHAP-guided counterfactual. For each building, we switch its largest positive SHAP driver with the most efficient alternative inferred from SHAP, re-estimate EUI, and record percent savings. Aggregating across buildings yields a




(a) SHAP beeswarm for Heating Sources.




(b) SHAP for Window-to-Wall Ratio.

Figure 3: SHAP explanation of two key predictors of building energy use: (a) Packaged Terminal Heat Pumps linked to higher EUI, and (b) Window-to-Wall Ratio exhibiting a U-shaped effect, where both low and high values raising energy demand.